Question 4 - assignment 05

1) $\hat{y} = b_0 + \sum_{k=1}^{K} b_k x_k$

$$\frac{\partial \hat{y}}{\partial x_k} = b_k$$

A 1 unit increase in x_k changes the prediction by exactly b_k. This is constant meaning it does not depend on x.

2) $\hat{p} = \frac{e^{b \cdot x}}{1 + e^{b \cdot x}}$

let $L = b \cdot x$

$$\hat{p} = \frac{e^L}{1 + e^L}$$

$$\frac{d}{dL}\left(\frac{e^L}{1+e^L}\right) = \frac{e^L}{(1+e^L)^2} = \frac{e^L}{1+e^L}\left(1 - \frac{e^L}{1+e^L}\right)$$

$$\frac{d\hat{p}}{dL} = \hat{p}(1-\hat{p})$$

Chain rule:

$L = b \cdot x \qquad \frac{\partial L}{\partial x_k} = b_k$

$$\frac{\partial \hat{p}}{\partial x_k} = \frac{d\hat{p}}{dL} \cdot \frac{\partial L}{\partial x_k} = \hat{p}(1-\hat{p}) \cdot b_k$$

$$\frac{\partial \hat{p}}{\partial x_k} = b_k \hat{p}(1-\hat{p})$$

Unlike linear regression, the effect of x_k on $\hat{p}$ is not constant because $\frac{\partial \hat{p}}{\partial x_k} = b_k \hat{p}(1-\hat{p})$ which depends on $\hat{p}$.

2 continued)

$\hat{p}(1-\hat{p})$ is maximized at $\hat{p} = 0.5$

$\hat{p}(1-\hat{p}) = 0.25$

$$\frac{\partial \hat{p}}{\partial x_k} \approx \frac{b_k}{4}$$

When $\hat{p} \approx 0.5$, the slope is the largest. A 1 unit increase in x_k changes the probability by approximately $\frac{b_k}{4}$ when $\hat{p}$ is near 0.5.

3) $\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = b \cdot x$

$$\frac{\partial}{\partial x_k}\left[\log\left(\frac{\hat{p}}{1-\hat{p}}\right)\right] = \frac{\partial(b \cdot x)}{\partial x_k} = b_k$$

A 1 unit increase in x_k increases the log-odds by b_k